

Regard $n \times n$ matrix A as a sequence $(C_1, C_2, \dots, C_n)$ of n col vectors in $\mathbb{R}^n$

Thm There exists a unique function $D: n \times n \text{ matrices} \rightarrow \mathbb{R}$ such that

- D is multilinear in cols of A (ie. linear as a function of i^{th} col when other cols are fixed)
- D is alternating (ie. $D(A) = 0$ if two adjacent cols of A are identical)
- $D(I) = u$, where u is a pre-specified real number (when $u=1$, we call $D = \text{determinant}$)

Proof Step 1 Claim: There is at most one such function D . We prove this by taking one such D and showing a forced formula for D . Eg. for $n=2$ we have $D \begin{bmatrix} a & b \\ c & d \end{bmatrix} = D(ae_1 + ce_2, be_1 + de_2)$

using multilinearity

$$ab \underbrace{D(\cancel{e_1}, e_1)}_0 + ad \underbrace{D(e_1, e_1)}_u + cb \underbrace{D(e_2, e_1)}_{=-D(e_1, e_2)=-u} + cd \underbrace{D(\cancel{e_2}, \cancel{e_2})}_0 = (ad - cb)u$$

For $n \times n$ A , the same procedure gives $D(A)$ using sum of n^n terms
multilinearity

using
alternating property

$$\sum_{\sigma \in S_n} a_{\sigma(1)1} \cdots a_{\sigma(n)n} D(e_{\sigma(1)}, \dots, e_{\sigma(n)})$$

a sum of $n!$ terms.

Note: $\{\sigma(1), \dots, \sigma(n)\} = \{1, 2, \dots, n\}$

Notation S_n = set of permutations of $\{1, 2, \dots, n\}$

$$= \left(\sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1)1} \cdots a_{\sigma(n)n} \right) u$$

where $\text{sgn}(\sigma) = +1$ or -1

obtained by fixing, for each $\sigma \in S_n$

one sequence of swaps to turn

sequence $\sigma(1), \dots, \sigma(n)$ into $1, 2, \dots, n$

This finishes step 1.

It also suggests how to prove existence: try to give an intrinsic description of $\text{sgn}(\sigma)$ (i.e. indep of seq of swaps) and then use the last exprⁿ as defⁿ of D .

Step 2 Define $D(A) := \left[\sum_{\sigma \in S_n} (-1)^{l(\sigma)} a_{\sigma(1)} \cdots a_{\sigma(n)} \right] u$, where

for a permutation $\sigma \in S_n$, $l(\sigma) = \#$ of inversions in σ i.e. $|\{(i,j) \mid i < j, \sigma(i) > \sigma(j)\}|$
(Ex: $l(\sigma) = \min \#$ of adjacent swaps reqd to convert $\sigma(1) \cdots \sigma(n)$ to $1, 2, \dots, n$)

Observe that this formula is multilinear and satisfies $D(I) = u$
(check)

Let's see why D is alternating.
Suppose slots $i, i+1$ are equal in A .

Pair $\left\{ \begin{array}{l} \dots, \sigma(i), \sigma(i+1), \dots = \sigma \\ \dots, \sigma(i+1), \sigma(i), \dots = \sigma' \end{array} \right\}$
with

Check Ex.

$\sigma' = \sigma \circ S_i$, where S_i = permutation that swaps $i, i+1$

We have $|l(\sigma) - l(\sigma')| = 1$ because $(i, i+1)$ is an inversion in exactly one of σ, σ' & other inversions "pair up"

Properties ① $\det(A) = \det(A^T)$ and therefore thm is true if we replace "col" by "row"

$$\det(A^T) = \sum_{\sigma \in S_n} (-1)^{l(\sigma)} a_{1\sigma(1)} \cdots a_{n\sigma(n)}$$

Reindex $\sigma^{-1} = \tau$

$$= \sum_{\tau \in S_n} (-1)^{l(\tau^{-1})} a_{\tau(1)} \cdots a_{\tau(n)}$$

and observe $l(\sigma) = l(\sigma^{-1})$

$$X = [c_1 \cdots c_n]$$

$$AX = [Ac_1 \cdots Ac_n]$$

$$\left\{ \begin{array}{l} i < j, \sigma(i) > \sigma(j) \\ \updownarrow \\ \sigma^{-1}(k) < \sigma^{-1}(l), k > l \end{array} \right.$$

④ Row opⁿ

	cR_i	$R_i \leftrightarrow R_j$	$R_i + aR_j$
effect on $\det(A)$	$c \det(A)$	$-\det(A)$	$\det(A)$
	(multilinear)	(alternating)	

Can calculate $\det(A)$ by row opⁿ

2 cases

- $RREF(A) = I$
- $RREF(A) = \text{row of 0's}$
 $\det \stackrel{?}{=} 0$

Discussion near end of class

③ A is invertible $\Rightarrow \text{RREF}(A) = I \Rightarrow \det(A) \neq 0$

A not invertible $\Rightarrow \text{RREF}(A)$ has a 0 row $\Rightarrow \det(A) = 0$

A square matrix is invertible
 $\Updownarrow$
 $\det(A) \neq 0$ very useful theoretically!

$$\bullet \det(A) = \sum_{\sigma \in S_n} (-1)^{l(\sigma)} a_{\sigma(1)1} \cdots a_{\sigma(n)n}$$

$$= \sum_{\tau \in S_n} (-1)^{l(\tau)} a_{1\tau(1)} \cdots a_{n\tau(n)}$$

Ex

equivalent to recursive formula
 developing det by a column

equivalent to recursive formula
 developing det by a row

② $\det(AX) = \det(A) \det(X)$ Pf. Regard LHS and RHS as functions of matrix X and see that both functions are multilinear, alternating in cols of X and take $I \rightarrow \det(A)$. Use uniqueness thm!

(Note: if cols of X are $C_1, \dots, C_n$ then ⑤ $A \text{adj}(A) = \det(A) I = \text{adj}(A) A$ Look up meaning of $\text{adj}(A)$.)

(Cols of AX are $AC_1, \dots, AC_n$ Extra $\rightarrow$ These are $2n^2$ equalities ② Shows $AB = I \Rightarrow \det(A) \det(B) = 1$
 Proof? Again use uniqueness! ⑤ Shows $\det(A)$ invertible in $\mathbb{R} \Rightarrow$ matrix A invertible!)